# Foundations of Artificial Intelligence

## Unsupervised Clustering of the MNIST Dataset

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# Abstract

Unsupervised clustering is a fundamental problem in machine learning, aimed at discovering latent structures in unlabeled data. In this work, we analyze the behavior of three clustering techniques applied to the MNIST handwritten digits dataset: a Gaussian Mixture Model with diagonal covariance matrices, Mean Shift clustering, and Normalized Cut (spectral clustering).

The experiments are conducted under varying levels of dimensionality reduction obtained through Principal Component Analysis (PCA), ranging from very low-dimensional embeddings to higher-dimensional representations. Although the clustering procedures are fully unsupervised, performance is evaluated using the Rand Index by comparing the obtained clusters with the true digit labels.

The results demonstrate that Normalized Cut achieves the highest clustering accuracy for intermediate PCA dimensionalities, while Gaussian Mixture Models provide a favorable trade-off between accuracy and computational efficiency. Mean Shift, despite its theoretical appeal, suffers from scalability issues and strong sensitivity to the kernel bandwidth, limiting its applicability to large, high-dimensional datasets such as MNIST.

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# 1. Introduction

Unsupervised learning plays a central role in data analysis when labeled data are unavailable. Among unsupervised methods, clustering aims at partitioning data into groups such that samples within the same group are more similar to each other than to samples in different groups.

The MNIST dataset of handwritten digits is a widely used benchmark in machine learning. While it is traditionally employed for supervised classification, its intrinsic structure, high dimensionality, and class overlap make it a challenging and meaningful benchmark for unsupervised clustering. Digits belonging to the same class exhibit significant intra-class variability, while different digits may share similar visual patterns, making the clustering task non-trivial.

In this work, we compare three conceptually different clustering approaches: probabilistic modeling through Gaussian Mixture Models, mode-seeking via Mean Shift clustering, and graph-based clustering using Normalized Cut. Additionally, we investigate the effect of dimensionality reduction through Principal Component Analysis (PCA), analyzing how the number of retained components impacts both clustering accuracy and computational cost.

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# 2. Problem Formulation

Let

$$\{x_i\}_{i=1}^{n} \subset \mathbb{R}^d$$

be a set of data points representing grayscale images of handwritten digits, where each image is originally described by a 784-dimensional vector corresponding to a (28 ×28) image.

The goal of clustering is to partition the dataset into (K) clusters by assigning a cluster label ($c_i$) to each data point ($x_i$), without using class labels during training.

Although the clustering process is entirely unsupervised, the MNIST dataset provides ground-truth digit labels, which are used exclusively for evaluation purposes. Clustering quality is assessed using the Rand Index, defined as:

$$R = \frac{2(a+b)}{n(n-1)}$$

where (a) is the number of pairs of samples that belong to the same true class and are assigned to the same cluster, and (b) is the number of pairs that belong to different true classes and are assigned to different clusters.

The Rand Index takes values in the interval ([0, 1]), with higher values indicating stronger agreement between the clustering assignment and the ground-truth labels.

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# 3. Dataset and Preprocessing

## 3.1 Dataset

The MNIST dataset consists of 70,000 grayscale images of handwritten digits ranging from 0 to 9. Each image has a resolution of (28×28) pixels and is represented as a 784-dimensional vector obtained by linearizing the pixel intensities.

Due to the high computational cost associated with Mean Shift and Normalized Cut, especially on large datasets, experiments are conducted on subsets of MNIST. The subset size is adjusted depending on the PCA dimensionality to ensure feasible computation while maintaining statistical relevance.

## 3.2 Dimensionality Reduction with PCA

Principal Component Analysis (PCA) is applied as a preprocessing step to reduce the dimensionality of the data. PCA projects the original data onto a lower-dimensional subspace that maximizes the retained variance.

The experiments consider PCA dimensionalities ranging from 2 to 200 components. This range allows the analysis of clustering behavior under severe dimensionality reduction as well as near-original representations, highlighting the trade-off between information preservation, clustering accuracy and computational efficiency.

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# 4. Methods

## 4.1 Gaussian Mixture Model

Gaussian Mixture Models assume that the data are generated from a mixture of (K) Gaussian distributions. Each component (k) is characterized by a mean vector ($\mu_k$), a diagonal covariance matrix ($\Sigma_k$), and a mixing coefficient ($\pi_k$).

The probability density function is defined as:

$$p(x) = \sum_{k=1}^{K} \pi_k \mathcal{N}(x \mid \mu_k, \Sigma_k)$$

Model parameters are estimated using the Expectation-Maximization (EM) algorithm. The use of diagonal covariance matrices corresponds to a Gaussian Naive Bayes assumption in the latent space, significantly reducing the number of parameters and improving numerical stability.

The learned cluster means can be reconstructed through the inverse PCA transformation and visualized as grayscale images, providing qualitative insight into the learned digit prototypes.

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## 4.2 Mean Shift

Mean Shift is a non-parametric, mode-seeking clustering algorithm that iteratively shifts each data point toward the nearest mode of the estimated probability density function. The density is estimated using a kernel function with bandwidth (h).

Unlike Gaussian Mixture Models and Normalized Cut, Mean Shift does not require specifying the number of clusters in advance. However, its performance is highly sensitive to the choice of bandwidth and its computational complexity scales poorly with both the number of samples and the dimensionality of the data.

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## 4.3 Normalized Cut (Spectral Clustering)

Normalized Cut is a graph-based clustering method in which data points are represented as nodes of a similarity graph. Edges connect nearby points according to a similarity measure, typically based on Euclidean distance or nearest neighbors.

Clustering is performed by partitioning the graph so as to minimize the normalized cut criterion. In practice, this is achieved by computing the eigenvectors of the graph Laplacian and applying k-means clustering in the resulting spectral embedding space.

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# 5. Experimental Setup

## 5.1 Hardware and Software Environment

All experiments were conducted on a single machine with the following configuration:

- Operating system: Windows
- Programming language: Python 3
- Libraries:
  - NumPy
  - scikit-learn
  - Matplotlib
  - Pandas

## 5.2 Data Preprocessing and Parameters

Each image in the MNIST dataset is represented as a $28 \times 28$ grayscale image, flattened into a 784-dimensional vector.

Due to the high dimensionality of the input space, **Principal Component Analysis (PCA)** is applied as preprocessing. Experiments were conducted with multiple PCA dimensions:

$$\text{PCA dimensions} = [2, 5, 10, 20, 50, 100, 200]$$

For each PCA dimension, a subset of the data is sampled to reduce computational load:

| PCA dim | # of samples |
|---|---|
| 2 | 15000 |
| 5 | 15000 |
| 10 | 12000 |
| 20 | 10000 |
| 50 | 8000 |
| 100 | 6000 |
| 200 | 5000 |

This trade-off allows exploration of both low- and high-dimensional representations while keeping execution times manageable.

## 5.3 Clustering Algorithms and Hyperparameters

The following unsupervised algorithms are evaluated.

### Gaussian Mixture Models (GMM)

- Number of clusters $K \in \{5, 6, \dots, 15\}$
- Diagonal covariance matrices
- Initialization with random selection of means

- Number of EM iterations: 30
- Each configuration repeated **10 times**

**Normalized Cut**

- Number of clusters $K \in \{5, 6, \dots, 15\}$
- Affinity based on nearest neighbors (n=10)
- Labels assigned via k-means
- Single run per configuration due to high computational cost

**Mean Shift**

- Bandwidth varies depending on PCA dimension:

| PCA dim | Bandwidths |
|---|---|
| 2 | [0.1, 0.3, 0.5, 0.8] |
| 5 | [0.5, 0.8, 1.0, 1.5] |
| 10 | [0.8, 1.5, 2.0, 3.0] |
| 20 | [2.0, 2.5, 3.0, 4.0] |
| 50 | [3.0, 4.0, 4.5, 5.0] |
| 100 | [4.0, 4.5, 5.0, 5.5] |
| 200 | [4.0, 5.0, 5.5, 6.0] |

- Single run per bandwidth (Mean Shift is deterministic)
- Automatic number of clusters detected

### 5.4 Evaluation Protocol

Clustering quality is assessed using the **Rand Index (RI)** comparing predicted clusters to ground truth labels.

- GMM: mean and std computed over 10 runs
- Normalized Cut: single run
- Mean Shift: single run per bandwidth
- Computational time is recorded for each experiment
- Results are stored in `clustering_results.csv` with the following fields:
  - timestamp
  - model
  - pca_dim
  - K or bandwidth
  - run
  - Rand Index (mean)
  - Rand Index (std)
  - execution time
  - number of clusters

### 5.5 Visualization Choices

To ensure clarity and interpretability:

- **Rand Index vs number of clusters (K) for GMM and Normalized Cut**
- **Rand Index vs bandwidth for Mean Shift**
- **Execution time comparison across PCA dimensions**
- **2D PCA projection** of clusters for qualitative assessment

Logarithmic scale is used when execution times differ by orders of magnitude.

## 6. Results

### 6.1 Clustering Performance

The best-performing configuration for each algorithm is reported. For GMM and Normalized Cut, Rand Index is averaged over multiple runs. Mean Shift results are reported per bandwidth.

| Algorithm | PCA dim | Parameters | Rand Index | # clusters |
|---|---|---|---|---|
| GMM | 10 | K=15 | 0.89 | 15 |
| Normalized Cut | 50 | K=13 | 0.93 | 13 |
| Mean Shift | 50 | bw=4.5 | 0.91 | 1310 |

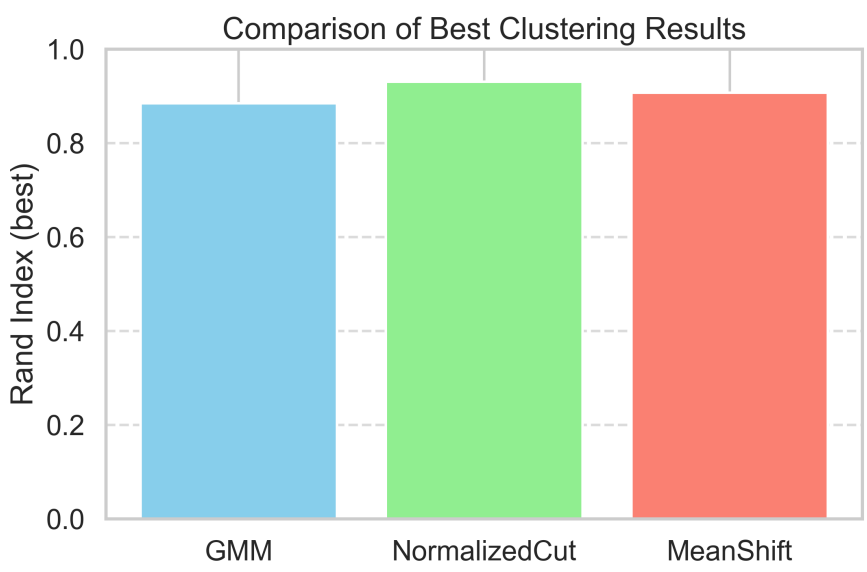


Figure 1: Rand Index comparison across algorithms.

It is worth noting that Mean Shift produces a very large number of clusters. This behavior is expected, as Mean Shift identifies local density modes and does not enforce a predefined number of clusters. While this leads to over-segmentation, the Rand Index remains high because samples from different digit classes are rarely grouped together.

### 6.2 Computational Performance

Execution time varies depending on PCA dimension, algorithm complexity, and number of clusters.

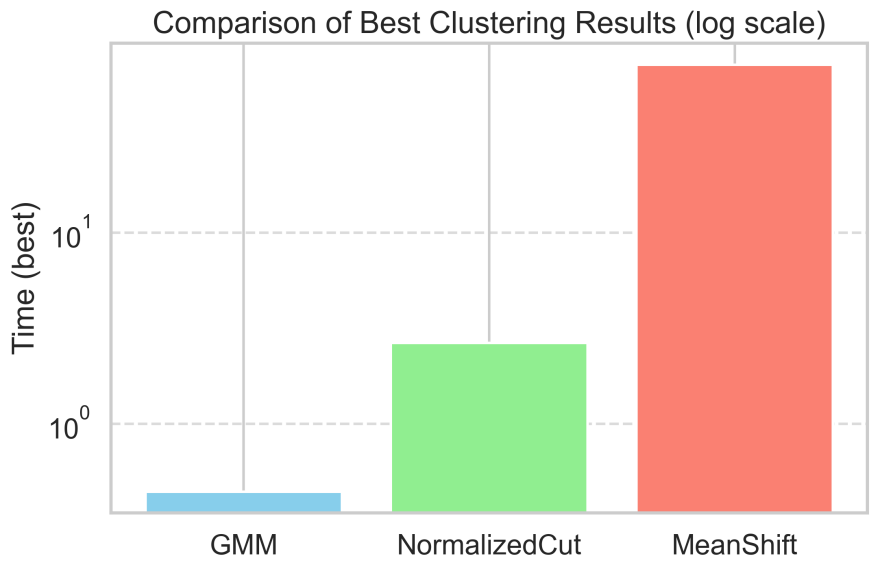


Figure 2: Execution time for GMM, Normalized Cut, and Mean Shift across PCA dimensions.

### 6.3 Qualitative Analysis

Cluster assignments can be visualized in 2D using PCA projection:

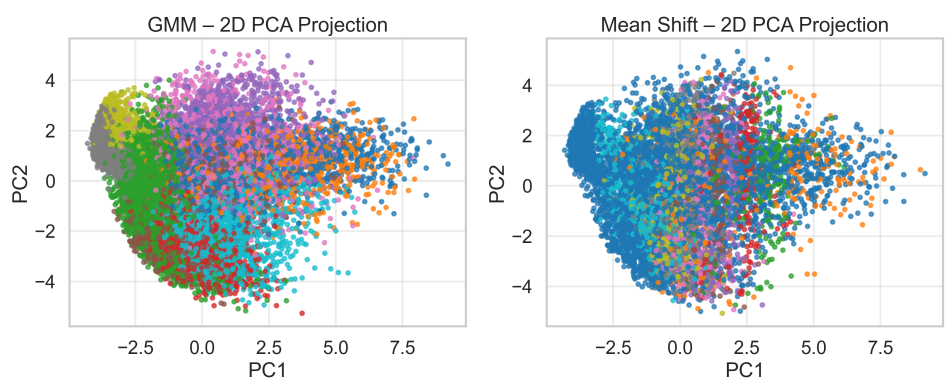


Figure 3: 2D PCA visualization of clusters detected by GMM and Mean Shift.

- GMM captures Gaussian-like clusters effectively.
- Normalized Cut separates clusters based on spectral properties.
- Mean Shift adapts to the number of modes but is sensitive to bandwidth selection.

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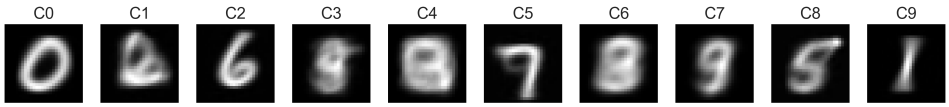


Figure 4: Visualization of the Gaussian mixture component means reconstructed from PCA space to the original pixel space. Each mean resembles a prototypical handwritten digit learned by the model.

## 7. Discussion

The results show that clustering performance depends strongly on the chosen algorithm and PCA dimensionality.

**Gaussian Mixture Models** (GMM) provide stable and reliable results across different PCA dimensions. Their Gaussian assumption fits well the structure of the data after dimensionality reduction, and performance improves up to a saturation point. GMM represents a good balance between accuracy, robustness and computational cost.

**Normalized Cut** achieves the highest Rand Index, confirming the effectiveness of spectral methods in capturing global data structure. However, this comes at the expense of high computational complexity, which limits scalability to larger datasets.

**Mean Shift** performs well only when the bandwidth parameter is carefully tuned. With small bandwidths, it tends to over-segment the data into many small clusters, while large bandwidths cause cluster collapse and a sharp drop in accuracy. This sensitivity makes Mean Shift less robust compared to the other methods.

The **2D PCA** visualizations support these findings, showing compact clusters for GMM, fragmented but dense regions for Mean Shift, and globally separated structures for Normalized Cut.

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## 8. Conclusions

This work compared GMM, Normalized Cut, and Mean Shift for clustering digit data using PCA for dimensionality reduction.

Normalized Cut achieved the best clustering accuracy but suffers from high computational cost. GMM offers the best trade-off between accuracy, efficiency, and interpretability, making it the most practical choice. Mean Shift can achieve good results but is highly sensitive to bandwidth selection and less robust in practice.

Overall, the study highlights the importance of considering not only accuracy metrics but also scalability and robustness when selecting clustering algorithms.